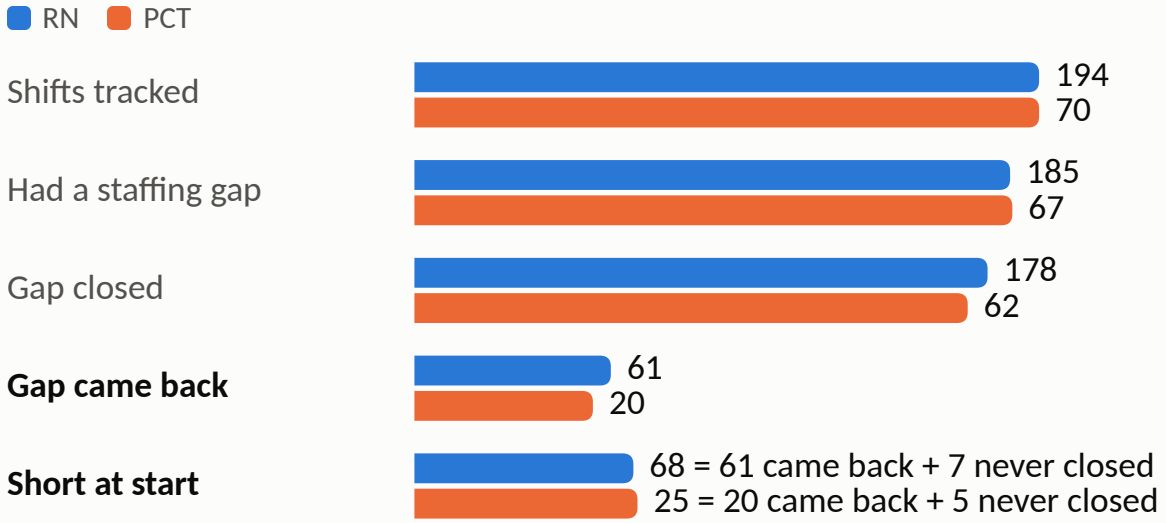
01 · FINDING

9 in 10 RN shifts that started short had already looked covered

Life of a gap shifts, by role



Example: Willow Medical, RN day shift staff by hours before start

	72h	60h	48h	36h	24h	12h	0h
Required	5	5	5	5	5	5	**6**
Available	4	5	5	5	5	5	5
Open need	**1**	**0**	0	0	0	0	**1**

- 60h: gap closes with exactly enough staff
- 0h: requirement rises, shift starts one RN short

RN

61 of 68

PCT

20 of 25

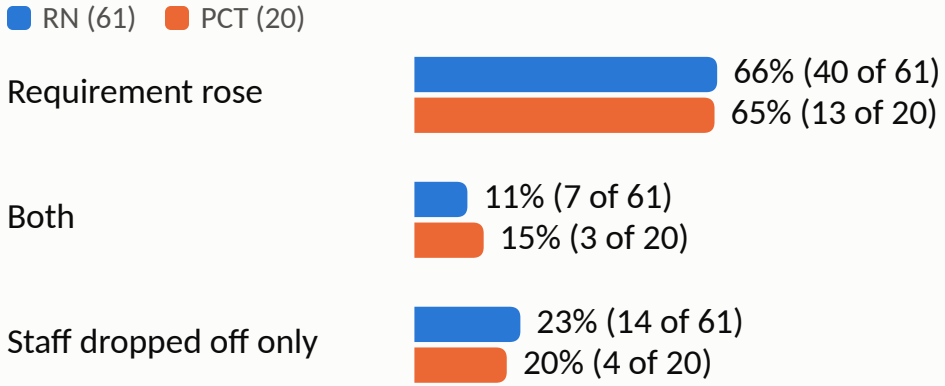
shifts that started short had a gap that was closed earlier, then reopened

Short at start = open need at the 0h snapshot (gaps that came back plus gaps that never closed). Source: synthetic staffing snapshots; RN and PCT kept separate.

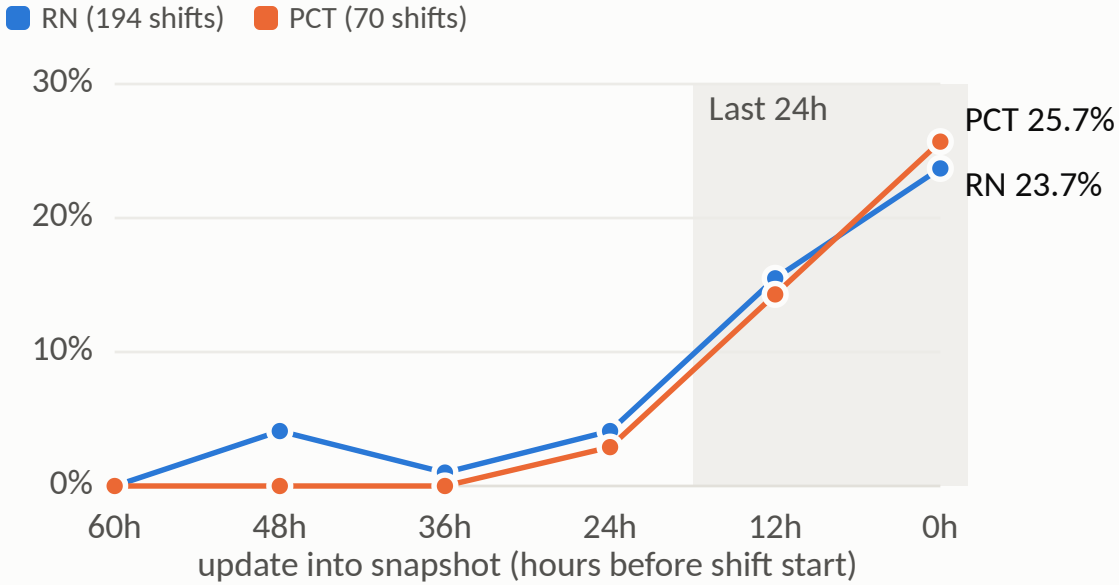
02 · EXPLANATION

Gaps reopen because the requirement rises, not because staff leave

What changed between closure and return % of returns



Increases cluster in the final 24 hours % of 12h updates where required rose



Ruled out: “paper” closures returning more (RN 7/21 vs 51/140) · early closures being fragile (an exposure effect) · ±1 noise (29 of 61 RN returns were 2 short)

Requirement rose in 47 of 61 RN and 16 of 20 PCT returns, counting “both”. Source: synthetic staffing snapshots; RN and PCT kept separate.

03 · RECOMMENDATION

Test why requirements rise late, and protect no-spare closures in the meantime

Why ask: gaps come back because the requirement rises late

47 of 61

RN gaps that came back were reopened by a requirement increase

% of 12-hour updates where the RN requirement rose all 194 RN shifts

0–4%	16%	24%
60h to 24h before start	into 12h before start	into shift start

The requirement barely moves for two days, then rises sharply in the final 24 hours.

RECOMMENDATION

Test what drives late requirement increases

1. Pull the forecast revision history (census, admissions, acuity) behind each late increase.
2. Check whether each increase was visible 24–48h earlier. If it was, forecast earlier; if not, hold a buffer.
3. Meanwhile, flag closures with no spare as “at risk” until shift start. After a rise, they came back 38 of 59 times, vs 9 of 35 with one spare.

PCT follows the same pattern: 16 of 20 returns followed an increase; 14% and 26% of updates rose at 12h and 0h. Limits: shows that the requirement rose, not why · 0h snapshot is a proxy · selected synthetic sample.